RESEARCH NOTEBOOK

Identical-Name Circuit Discovery in GPT-2 Medium

Research Area: Mechanistic Interpretability, Transformer Circuits, Entity Disambiguation

Model: GPT-2 Medium

Model Parameters: Approximately 355 million

Session: 01

Date: 7 November 2026

Status: Initial experiments completed; causal investigation pending.

1. Research Objective

The objective of this project is to investigate how transformer language models process identical names that may refer to different real-world entities.

The central example is:

"John gave a bag to John."

The two occurrences of "John" share the same lexical name but occupy different semantic roles.

John 1: Giver.

John 2: Receiver.

The underlying research question is:

At which stages of a transformer network does information emerge that distinguishes two occurrences of the same name, and which internal computational circuits contribute to that distinction?

The broader ambition is to investigate whether internal transformer circuits can contribute to entity disambiguation rather than merely detecting lexical repetition.

2. Research Hypotheses

The initial investigation considers four hypotheses.

H1 — Contextual differentiation

Two occurrences of the same name develop different contextual representations as they pass through transformer layers.

H2 — Repetition-sensitive attention

Some attention heads respond differently when two positions contain the same lexical name compared with different names.

H3 — Semantic role representation

Transformer components encode distinctions between grammatical roles, such as giver and receiver.

H4 — Entity disambiguation

A collection of internal computational components may contribute to distinguishing two separate entities that share the same name.

H4 is the principal long-term research objective. It has not yet been demonstrated.

3. Experimental Environment

The selected model is GPT-2 Medium.

Architecture property	Value
Transformer layers	24
Attention heads per layer	16
Hidden dimension	1,024
Parameters	Approximately 355 million

Libraries:

PyTorch

Transformers

TransformerLens

The investigation uses internal activations, attention patterns, cosine similarity and, in later experiments, activation patching.

4. Experiment 1 — Token Representation
4. 4.1 Input

"John gave a bag to John."

4.2 Tokenization output

python

tensor([[50256, 7554, 2921, 257,

6131, 284, 1757, 13]])

Token positions:

Position	Token ID	Token
0	50256	Special beginning token
1	7554	John
2	2921	gave
3	257	a
4	6131	bag
5	284	to
6	1757	John
7	13	.

Observation:

The two occurrences of John have different token IDs because GPT-2 uses whitespace-sensitive byte-pair tokenization.

The first occurrence is represented by token ID 7554, while the second is represented by token ID 1757.

This demonstrates that identical written names do not necessarily correspond to identical token IDs.

Importantly, the two occurrences remain lexically related.

5. Experiment 2 — Residual Stream Representation Analysis
5. 5.1 Objective

Investigate how the representations of the two John positions evolve across transformer layers.

The metric selected was cosine similarity.

The formula:

$$\text{cosine_similarity}(A,B) = (A \cdot B) / (||A|| \times ||B||)$$

5.2 Initial results

Layer	Cosine similarity
0	0.975937
3	0.959708
6	0.940065
9	0.908481
12	0.886895
15	0.892313
18	0.914004
21	0.947311
23	0.954089

5.3 Observations

Three broad phases can be identified.

Early layers (0–9):

Similarity decreases progressively.

Middle layers (9–12):

The representations reach their lowest measured similarity.

Later layers (12–23):

Similarity increases again.

The lowest measured similarity is 0.886895 at layer 12.

The final measured similarity is 0.954089 at layer 23.

5.4 Preliminary interpretation

The observed non-monotonic behaviour suggests that contextual representations evolve throughout the network.

One possible explanation is that intermediate layers encode different contextual or grammatical features, while later layers develop representations with greater shared lexical or semantic information.

This remains a hypothesis.

Cosine similarity alone cannot establish entity identity, grammatical-role understanding or semantic equivalence.

6. Experiment 3 — Attention Head Discovery
6. 6.1 Objective

Identify attention heads that direct attention from the second occurrence of John towards the first occurrence.

The investigated attention component is:

python

blocks.{layer}.attn.hook_pattern

The measured attention value is:

python

attention[0, head, 6, 1]

Here:

Query position = 6 (second John).

Key position = 1 (first John).

6.2 Initial candidate heads

The strongest attention connections were:

Layer	Head	Attention
1	4	0.417514
6	4	0.378001
7	11	0.367523
23	5	0.290854
0	5	0.288782
0	10	0.265607
4	6	0.223009
22	2	0.185378
2	11	0.177442

21 7 0.177325

Four heads were selected for further investigation:

L1H4

L6H4

L7H11

L23H5

6.3 Attention distribution analysis

The attention distribution from the second John was examined.

Layer 1, Head 4

Position Token Attention

0 Beginning token 0.12575

1 John 0.41751

2 gave 0.00152

3 a 0.01597

4 bag 0.00038

5 to 0.01306

6 John 0.42580

Observation:

L1H4 allocates most of its attention to the two John positions.

This makes it an interesting candidate for investigating repeated-name sensitivity.

Layer 6, Head 4

Position Token Attention

0 Beginning token 0.58046

1 John 0.37800

6 John 0.03994

Observation:

This head allocates considerable attention to the beginning-of-sequence position and the first John.

Layer 7, Head 11

Position	Token	Attention
0	Beginning token	0.37568
1	John	0.36752
6	John	0.25671

Observation:

This head distributes attention across the beginning token, first John and second John.

Layer 23, Head 5

Position	Token	Attention
0	Beginning token	0.00010
1	John	0.29085
2	gave	0.18964
3	a	0.07281
4	bag	0.06064
5	to	0.28519
6	John	0.10076

Observation:

L23H5 attends to several parts of the sentence, particularly the first John and the preposition "to".

It may be worth investigating as a candidate for broader contextual integration.

7. Experiment 4 — Controlled Repeated-Name Comparison
7. 7.1 Objective

Determine whether the selected attention heads behave differently when the subject and recipient have identical or different names.

Three controlled sentences were tested:

S1: John gave a bag to John.

S2: John gave a bag to Michael.

S3: Michael gave a bag to John.

The measurement was attention from the recipient position towards the subject position.

7.2 Results

Attention head	S1	S2	S3
L1H4	0.41751	0.01892	0.01076
L6H4	0.37800	0.00090	0.00277
L7H11	0.36752	0.00001	0.00000
L23H5	0.29085	0.26618	0.26483

7.3 Main observation

L1H4 demonstrates a substantial change in attention behaviour when the subject and recipient names match.

The attention to the subject position is approximately:

0.41751 in the repeated-name condition.

0.01892 in the John-to-Michael condition.

0.01076 in the Michael-to-John condition.

This is consistent with sensitivity to repeated-name conditions.

L6H4 and L7H11 also exhibit similar behaviour.

L23H5, in contrast, maintains relatively stable attention values across the three conditions.

7.4 Preliminary conclusion

L1H4 is currently our strongest candidate for investigating repeated-name-sensitive attention.

However, we have not demonstrated that it performs entity matching or entity disambiguation.

Alternative explanations include:

Lexical repetition detection.

Token embedding similarity.

Contextual pattern recognition.

Grammatical or positional effects.

Further controlled experiments are required.

8. Current Research Findings

The following findings have been established experimentally:

The two occurrences of John have different GPT-2 token IDs.

Their residual-stream representations have different cosine similarities across transformer depth.

Their representations become less similar in intermediate layers and more similar in later layers.

Several attention heads show strong attention from the second John towards the first.

L1H4 exhibits substantially higher attention to the subject position when the subject and recipient names match.

L23H5 displays comparatively stable attention across the three tested sentences.

The central unresolved question is whether these attention patterns contribute causally to entity-related processing.

9. Important Scientific Limitations

The current investigation has several limitations.

Identical names do not necessarily indicate identical entities.

Two people can share the same name.

Different names do not necessarily indicate different entities.

An individual can be referenced by different names, aliases or descriptions.

Attention weights are not proof of computational function.

A head attending to a token does not establish what information it transmits or whether that information is causally important.

Cosine similarity is not an identity metric.

Similar representations do not necessarily indicate identical entities.

The experiment currently uses a small number of sentence templates.

Broader testing is necessary to establish generalisation.

10. Next Research Session — Experimental Plan

The next session will investigate whether L1H4 generalises beyond the name John.

Experiment 5 — Generalised Name Repetition

Test the following sentences:

John gave a bag to John.

Alice gave a bag to Alice.

Michael gave a bag to Michael.

John gave a bag to Michael.

Michael gave a bag to John.

Measure L1H4 attention from the recipient position towards the subject position.

Objective:

Determine whether L1H4 responds to repeated names generally rather than a particular lexical identity.

Experiment 6 — Explicit Entity Separation

Test:

"John gave a bag to another person named John."

Objective:

Investigate whether GPT-2 distinguishes repeated lexical names when the text explicitly describes different people.

Experiment 7 — Activation Patching

Use controlled model runs to intervene on candidate attention-head activations.

Initial candidate:

L1H4.

Objective:

Determine whether the candidate head causally contributes to a measurable model behaviour.

Experiment 8 — Circuit Reconstruction

Investigate the interactions between:

Attention heads.

MLP components.

Residual-stream representations.

Information propagated across transformer layers.

Objective:

Identify a candidate computational circuit associated with repeated-name processing and entity disambiguation.

11. Long-Term Research Objective

The ultimate objective is to investigate whether a relatively small collection of transformer components contributes to distinguishing entities that share identical names.

The desired outcome is a causally supported computational explanation rather than an attention visualisation alone.

The research will distinguish three separate concepts:

Lexical identity: Are the names the same?

Mention and role identity: How are the occurrences positioned and related?

Real-world entity identity: Do the mentions refer to the same individual?

These distinctions form the foundation of the project.

SESSION 1 SUMMARY

Model: GPT-2 Medium

Most promising candidate: Layer 1, Head 4.

Main observation: Strong repeated-name sensitivity in the tested attention pattern.

Most interesting representation result: Non-monotonic cosine similarity across transformer layers.

Next experiment: Test repeated names across John, Alice and Michael.

Current status: Preliminary observations obtained. Causal circuit discovery and entity disambiguation remain unproven.

Continuation instruction:

Resume from Experiment 5: Generalised Name Repetition.

Continue using GPT-2 Medium, retaining the previous experimental setup and measurements.

End of Session 1.